

Data Science

CSE558

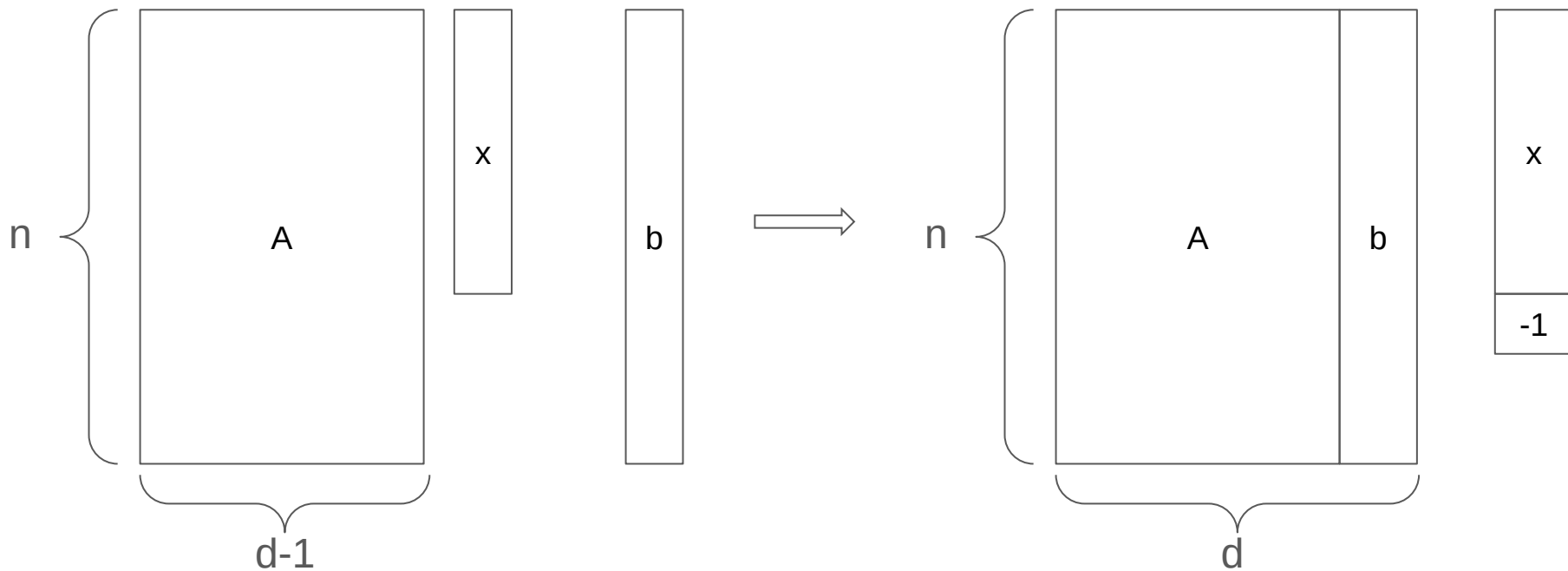
Recap

- Structured Low rank Approximation
 - CUR Decomposition
 - Matrix Completion.

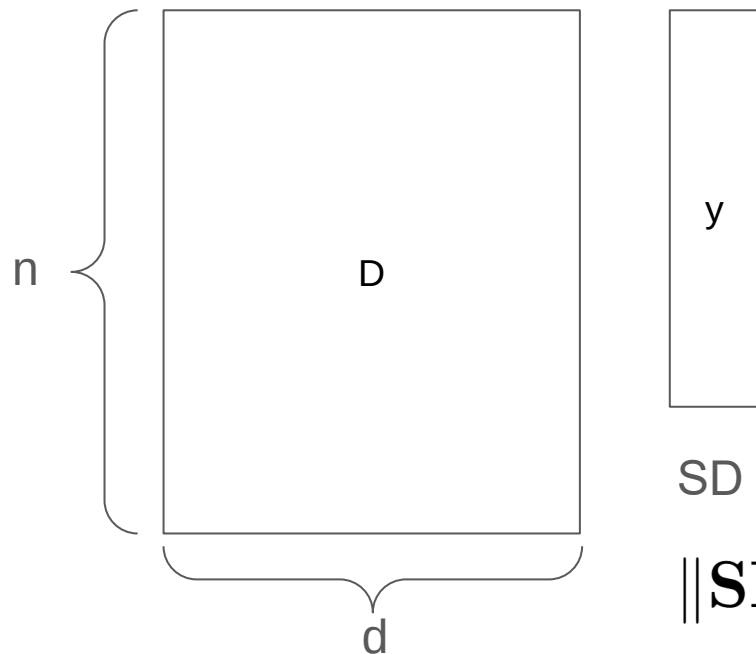
Overview

- Coresets
- Fast Linear Regression using Sampling
- Fast Clustering using Sampling

Linear Regression



Coresets For Linear Regression



$$D = [A; b] \text{ \& } y = [x; -1]$$

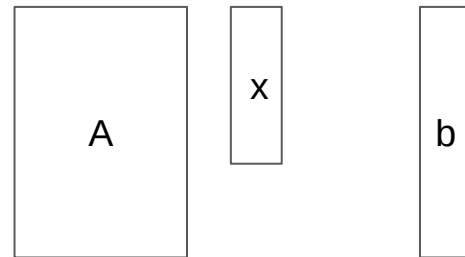
SD is ε -coreset of D if for every y ,

$$\|\mathbf{SDy}\|_2^2 \in (1 \pm \varepsilon) \|\mathbf{Dy}\|_2^2$$

Linear Regression

Let,

$$\mathbf{x}^* \in \operatorname{argmin}_x \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2$$
$$\hat{\mathbf{x}} \in \operatorname{argmin}_x \|\mathbf{S}\mathbf{A}\mathbf{x} - \mathbf{S}\mathbf{b}\|_2^2$$



$$\|\mathbf{S}\mathbf{D}\hat{\mathbf{y}}\|_2^2 \in (1 \pm \varepsilon) \|\mathbf{D}\hat{\mathbf{y}}\|_2^2 \quad \|\mathbf{S}\mathbf{D}\mathbf{y}^*\|_2^2 \in (1 \pm \varepsilon) \|\mathbf{D}\mathbf{y}^*\|_2^2$$

$\hat{\mathbf{y}} = [\hat{\mathbf{x}}; -1]$ $\mathbf{y}^* = [\mathbf{x}^*; -1]$

$$\|\mathbf{D}\hat{\mathbf{y}}\|_2^2 \leq (1 + 3\varepsilon) \|\mathbf{D}\mathbf{y}^*\|_2^2$$

$$\|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\|_2^2 \leq (1 + 3\varepsilon) \|\mathbf{A}\mathbf{x}^* - \mathbf{b}\|_2^2$$

$$(1-\varepsilon)\|Dg\|_2^2 \leq \|SDg\|_2^2 \leq \|SDy^*\|_2^2 \leq (1+\varepsilon)\|Dy^*\|_2^2$$

$$\therefore \|Dg\|_2^2 \leq \frac{1+\varepsilon}{1-\varepsilon} \|Dy^*\|_2^2$$

$$\text{for } \varepsilon \in (0, 1/3) \quad \frac{1+\varepsilon}{1-\varepsilon} = 1 + \frac{2\varepsilon}{1-\varepsilon} \leq 1 + \frac{2\varepsilon}{2/3} \leq 1+3\varepsilon$$

$$\therefore \|Dg\|_2^2 \leq (1+3\varepsilon)\|Dy^*\|_2^2 \quad \text{for } \varepsilon \in (0, 1/3)$$

Linear Regression

Require guarantee for every $\mathbf{x} \in \mathbb{R}^{d-1}$.

Each $\mathbf{x} \in \mathbb{R}^d$ has a $\mathbf{y} \in \mathbb{R}^d$.

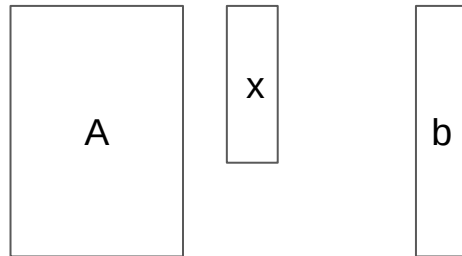
For every such $\mathbf{y} \in \mathbb{R}^d$ we require,

$$\|\mathbf{S}\mathbf{D}\mathbf{y}\|_2^2 \in (1 \pm \varepsilon)\|\mathbf{D}\mathbf{y}\|_2^2$$

We take union bound over all $\mathbf{x} \in \mathbb{R}^{d-1}$.

There are infinite many such $\mathbf{x}$'s.

ε -net argument.



ϵ -Net

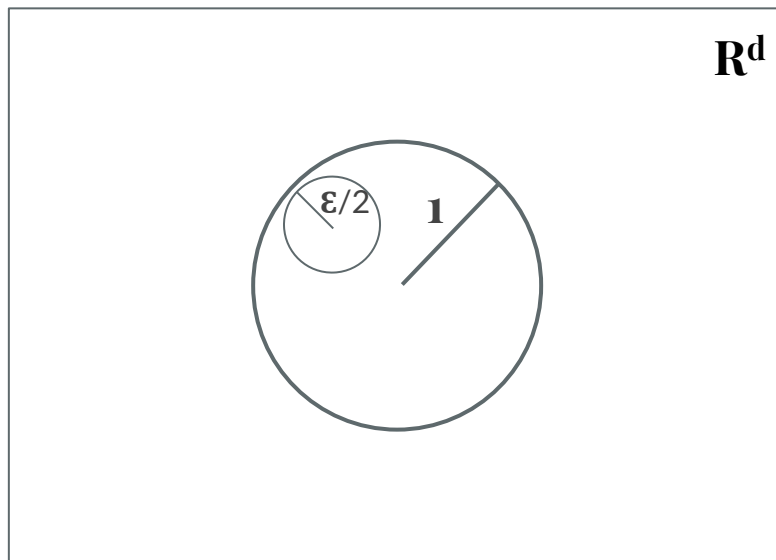
For the set with all unit vectors $\mathbf{y} \in \mathbb{R}^d$, a set $\mathcal{N}$ is said to be its ϵ -net, if for every $\mathbf{y}$, there is a $\mathbf{w} \in \mathcal{N}$ such that, $\|\mathbf{y} - \mathbf{w}\| < \epsilon$.

$$\|\mathbf{S}\mathbf{D}\mathbf{w}\|_2^2 \in (1 \pm \epsilon) \|\mathbf{D}\mathbf{w}\|_2^2$$

$$\|\mathbf{S}\mathbf{D}\mathbf{y}\|_2^2 \in (1 \pm O(\epsilon)) \|\mathbf{D}\mathbf{y}\|_2^2$$

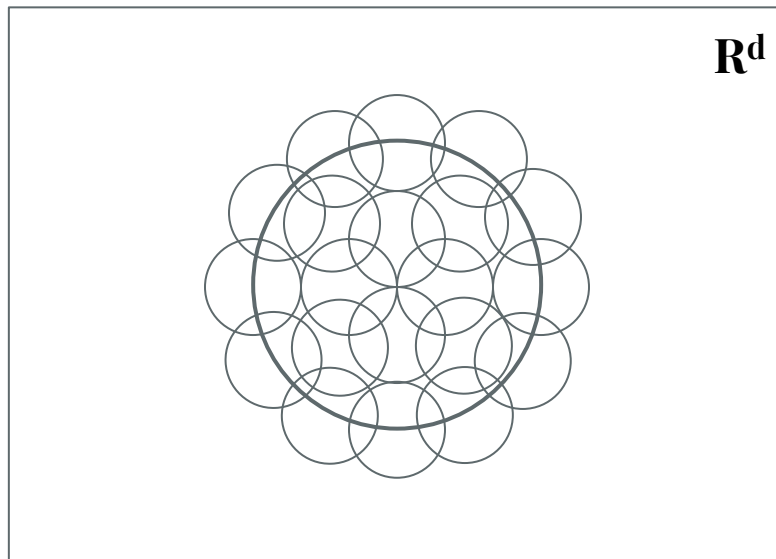
ϵ -Net

Consider a unit sphere in d-dimensional space,
This sphere consists of every $\mathbf{y} \in \mathbb{R}^d$



$$\text{Volume} = 1^d = 1$$

ε -Net



Volume of $\varepsilon/2$ -balls: $(\varepsilon/2)^d$.

So $\#(\varepsilon/2\text{-balls})$ is $1/(\varepsilon/2)^d = (\varepsilon/2)^{-d}$

Recap

Coreset

- Epsilon Net

Overview

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 - Epsilon Net
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ϵ -Net

Let $M = \{z \in \mathbb{R}^n \mid z = Dy \text{ for some } y \in \mathbb{R}^d \text{ and } \|z\|_2 = 1\}$

Let N be $\frac{1}{2}$ -net of M . We know $|N| = O(2^d)$

Let $z = z^0 + z^1 + z^2 + \dots$

$$D: y \rightarrow z$$

$$D^T: z \rightarrow y$$

$$D^T: z \rightarrow w$$

for $z \in N$

$$\text{SD is } \epsilon\text{-coreset for set } N \Rightarrow \|SDw\|_2^2 \in (1 \pm \epsilon) \|Dw\|_2^2$$

$\|z^i\| \leq \frac{1}{2^i}$ & z^i is vector in N with some scalar multiplication.

$$\|z - z^0\|_2^2 \leq \frac{1}{2} \quad ; \quad z = z^0 + (z - z^0)$$

ϵ -Net Analysis

$z - z' = z' + (z - z'' - z')$, z' is a vector in N multiplied with a scalar.

$$\| \underline{z - z'' - z'} \|$$

$$\tilde{z} = z - z'', \|\tilde{z}\| \leq \frac{1}{2}, \quad \hat{z} = \frac{\tilde{z}}{\|\tilde{z}\|}, \quad \hat{z}_i \in N \text{ s.t. } \|\hat{z} - \hat{z}'\|_2^2 \leq \frac{1}{2}$$

$$\therefore \|\tilde{z} - z'\| \leq \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$z = z'' + z' + z^2 + \dots$$

$$+ (z - z'' + z' + z^2 + \dots)$$

$$S D y = S z = S(z^0 + z^1 + z^2 + \dots)$$

$$\begin{aligned} \therefore \|S z\|_2^2 &= \|S(z^0 + z^1 + \dots)\|_2^2 \\ &= \sum_{0 \leq i < j < \infty} \|S z^i\|_2^2 + 2 \langle S z^i, S z^j \rangle \end{aligned}$$

$$= \left(\sum_{0 \leq i < j < \infty} \|z^i\|_2^2 + 2 \langle z^i, z^j \rangle \right) \pm 2\varepsilon \sum_{0 \leq i < j < \infty} \|z^i\| \|z^j\|$$

$$= \|z\|_2^2 \pm O(\varepsilon) \|z\|_2^2$$

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 - Epsilon Net
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Coreset for Linear Regression

$$\|\mathbf{SDy}\|_2^2 \in (1 \pm \varepsilon) \|\mathbf{Dy}\|_2^2 \quad \text{a.k.a. subspace embedding.}$$

For a fixed vector y consider the random variable

$$r = \left\{ \frac{(D_i^T y)^2}{p_i}, \quad \text{w.p. } p_i \right.$$

$\mathbb{E}[r] = \sum_{i=1}^n \frac{(D_i^T y)^2}{p_i} p_i = \|Dy\|_2^2$

Take m iid samples from r , say $r_1, r_2, \dots, r_m$.

Return average. $R = (r_1 + r_2 + \dots + r_m)/m$.

Check for unbiasedness and realize the probability distribution.

$$\text{Var}(R) = \frac{1}{m^2} \sum_{i=1}^m \text{Var}(r_i) = \frac{1}{m^2} \sum_{i=1}^m \left[\sum_{j=1}^n \left(\frac{(D_j^T y)^2}{p_j} \right)^2 \cdot p_j - \|Dy\|_2^4 \right]$$

let $p_j = \frac{(D_j^T y)^2}{\|Dy\|_2^2}$ then $\downarrow$

$$\frac{1}{m^2} \sum_{i=1}^m \left[\sum_{j=1}^n \frac{(D_j^T y)^4}{\cancel{(D_j^T y)^2}} \|Dy\|_2^2 - \|Dy\|_2^4 \right]$$

$$= \frac{1}{m^2} \sum_{i=1}^m \left(\cancel{\|Dy\|_2^4} - \cancel{\|Dy\|_2^4} \right) = 0$$

Coreset for Linear Regression

$$E[R] = \|Dy\|_2^2$$

Scaling ensures unbiasedness.

Distribution?

Analyze $\text{Var}(R)$.

Sensitivity:

Property: $\sup_{y \in \mathbb{R}^{d+1}} \frac{(D_i^T y)^2}{\|Dy\|_2^2} = s_i$

$$= \sup_{y \in \mathbb{R}^d} \frac{(D_i^T y)^2}{\sum_{j=1}^n (D_j^T y)^2}$$

Sensitivity Analysis

$$\sup_y \frac{(D_i^T y)^2}{\|Dy\|_2^2} \quad \text{for any } y \in \mathbb{R}^d \quad \frac{(D_i^T y)}{\|Dy\|_2^2} \leq \boxed{\quad}$$

independent of y .

$$\begin{aligned} \frac{(D_i^T y)}{\|Dy\|_2^2} &= (D_i^T y) (y^T D^T D y)^{-1} (y^T D_i) \\ &= (D_i^T y) (Dy)^+ (y^T D^T)^+ (y^T D_i) \\ &= (D_i^T y) (y^+ D^+ (D^T)^+ (y^T)^+ y^T D_i) \end{aligned}$$

$$\begin{aligned} \|Dy\|_2^2 &= y^T D^T D y \\ (v^T w)^T &= w^T (v^T)^+ \end{aligned}$$

let $v \in \mathbb{R}^d$ then $VV^T \preceq I$ i.e. for every $x \in \mathbb{R}^d$

$$x^T VV^T x \leq x^T I x$$

there, $VV^T \in \mathbb{R}^{d \times d}$

$$V^T V = I$$

$$\therefore D_i^T (Y Y^T) D_i (D^T)^T (Y^T)^T Y^T D_i$$

$$\Rightarrow D_i^T (D^T D)^T D_i = S_i$$

Coreset for Linear Regression

Define $p_i = \frac{s_i}{\sum_{j \in [n]} s_j}$

For a fixed vector y consider the random variable

$$r = \left\{ \frac{(D_i^T y)^2}{p_i}, \quad \text{w.p. } p_i \right.$$

Take m iid samples from r , say $r_1, r_2, \dots, r_m$.

Return average. $R = (r_1 + r_2 + \dots + r_m)/m$.

Unbiasedness will hold!

$$\|\mathbf{SDy}\|_2^2 \in (1 \pm \varepsilon) \|\mathbf{Dy}\|_2^2$$

What is the smallest coreset size required to guarantee

Coreset Size

Bernstein's Inequality: Let $X_1, X_2, \dots, X_n$, be n independent scalar variables such that $|X_i - \mathbf{E}[X_i]| \leq \mathbf{b}$. Let their sum be $\mathbf{S} = X_1 + X_2 + \dots + X_n$ and $\sigma^2 = \mathbf{Var}[\mathbf{S}]$. Then for all $\mathbf{t} > 0$,

$$\mathbf{P} (|\mathbf{S} - \mathbf{E}[\mathbf{S}]| > \mathbf{t}) \leq 2 \cdot \exp \left(-\frac{2\mathbf{t}^2}{\sigma^2 + \mathbf{b}\mathbf{t}/3} \right)$$